

Botlien

Your robots are working. Is your business?

Financial telemetry for RaaS operators: robot telemetry turned into cash flow analysis, cash flow projections, default risk, and quota tracking, built directly for the operator running the fleet.

POSITIONING PIVOT 2026-08-05 — RAAS OPERATOR IS NOW THE PRIMARY CUSTOMER

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Executive summary

A RaaS operator's fleet and their financials run on two different clocks. The fleet reports task completions and uptime, if it reports anything at all. The business reports revenue and cash flow a month later, after invoices settle. Botlien connects the two, in real time.

Botlien turns a RaaS operator's own robot telemetry into financial telemetry, then quantifies it into the numbers that actually run the business: cash flow analysis, cash flow projections, default/credit risk on their own fleet, and whether they're hitting the monthly quota of whatever contract the RaaS line was built around. **The RaaS operator is the primary, direct customer** as of the 2026-08-05 positioning pivot, not a lender. Lenders remain relevant as a downstream expansion, sold once an operator has real telemetry history, not the entry point.

This is not a new financial concept. Fleet financial analytics and asset-backed lending are both decades-old categories. What's missing is the same real-time, cross-brand visibility layer vehicles already have through telematics (65–85% adoption in financed fleets), never built for robots, and never turned into a financial-planning tool for the operator itself.

RAISING \$500K pre-seed, milestone-framed: one or more signed RaaS-operator pilots generating first monitored-robot volume and a real case study, the trigger to raise a seed round.	WHERE THE COMPANY ACTUALLY IS Pre-product, pre-revenue, in active discovery led by the RaaS-operator track. Stated plainly throughout this document, not oversold.
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The problem & why now

RaaS operators run their fleets and their financials on two different clocks. Nothing today connects robot-level performance to business-level cash flow in real time, so an operator finds out they're short of quota, burning cash, or carrying an underperforming unit only when it already shows up on a bank statement, not when there's still time to act on it.

Three things true today that weren't two years ago

- Robotics-as-a-Service is growing at roughly 20% CAGR through 2030, roughly double the growth rate of the broader equipment finance industry it depends on for capital, meaning more operators, faster, all running this exact blind spot.
- Telemetry-based fleet monitoring is not a novel concept, it's mature and sometimes-mandatory in adjacent equipment classes. Adoption exceeds 85% for vehicles financed through major transportation lenders. What doesn't exist yet is that same telemetry turned into *financial* telemetry, useful to the operator running the business.
- An ISO mixed-fleet telematics standard was approved in November 2025, explicitly framed around enabling usage-based valuation, proof the underlying data category is now standardized enough to build a real financial product on top of.

THE FRAMING THAT MATTERS FOR INVESTORS

This isn't "should a business know its own cash flow," an already-answered question for every other business. It's "why is a RaaS operator's fleet data and financial data still living in two different places."

The solution

Botlien turns a RaaS operator's robot telemetry into financial telemetry, then quantifies it into the numbers that run the business.

The pipeline

- **Robot telemetry** — task completions, uptime, utilization, fault codes, pulled directly from the manufacturer's API.
- **Financial telemetry** — the same signals converted into realized revenue per robot, measured against that unit's actual contract or quota target.
- **Cash flow analysis and projections** — current-period actuals and a forward-looking view, so a shortfall is visible weeks before it hits a bank statement, not after.
- **Default / credit risk** — for operators who've financed their own fleet, or who extend financing to their own downstream customers, the same risk engine applies one layer earlier than a traditional lender would use it.

Once an operator has real telemetry history, a downstream product exists too: offering that verified data to their own upstream lender, the traditional collateral-monitoring pitch, now an expansion layer instead of the entry point.

Competitive landscape

Researched directly, twice, 2026-08-05, first via search then via a direct-fetch verification pass on both companies' own pages. Real white space exists, but it's narrower than "nobody's building this" — the two closest comps are named directly below rather than omitted.

CAPABILITY	BOTLIEN	FINDUSTRIAL	FORMANT
Financial telemetry (task completion → revenue)	Yes	Partial	Partial
Cash flow analysis (current period)	Yes	Yes	No
Cash flow projections (forward-looking)	Yes	No	No
Default/credit risk scoring	Yes	No	No
Quota/KPI tracking vs. contract	Yes	No	No
Robotics-specific	Yes	No	Yes
Sold to RaaS operators directly	Yes	No, sells to OEMs	Yes
Cross-brand normalization	Yes	No, per-OEM	Yes

Findustrial

75+ completed projects, 20+ logos including Agilox, Palfinger, Volvo, and Siemens Energy, EU co-funded. Targets OEM manufacturers transitioning to Equipment-as-a-Service, plus their financing partners, not RaaS operators directly. Real "CFO cockpit" for cash/margin tracking, but asset-agnostic (robotics, machinery, industrial vehicles) and confirmed by their own documentation to have zero predictive cash-flow projection or credit-risk scoring.

Formant

\$21M raised, past coverage cites 500%+ YoY revenue growth in 2023, customers including Blue River/John Deere, BP, and Knightscope. Built real cross-brand, operator-facing ROI reporting ("Formant Analytics"). But their current homepage has repositioned around "Formant Metaphysics," an AI incident-management platform, financial ROI reporting now looks secondary to their present pitch, if not phased out, which widens rather than narrows Botlien's gap.

PLAIN-LANGUAGE VERDICT

Neither company combines robotics-specific + operator-facing + predictive cash-flow + credit-risk scoring in one product. That four-part combination is Botlien's real, confirmed-open gap, verified on a second, direct-fetch pass, not just a first search-level read. Cendex Group, the original lender-facing comp, is re-confirmed to still be lender-only with no pivot toward operators.

Technical architecture

Four layers, each intentionally small in scope for a two-person team.

1 — Robot connectors

Starting brand: Bear Robotics — fully public gRPC/REST schema, no partner application needed just to read it. Streams connection state, battery, e-stop, mission, pose, error codes with severity, stuck-state detection, and velocity in one subscription. Pudu is real but gated behind a developer application, brand #2. Humanoid robots (Unitree, Agility Robotics) are a separate, less mature category on public fleet APIs, relevant because of the IntBot lead but not the technical proof point.

2 — Ingestion & normalization layer

A common schema mapping every brand's data into one "robot health record." VDA 5050 and MassRobotics AMR Interoperability already standardize position/battery/status across brands, normalizing rides an existing trend. A separate data-quality/pipe-health layer distinguishes "the robot went dark" from "our connection to the manufacturer's cloud went dark."

3 — Risk scoring engine

Rules-based at first, not machine learning. PD (probability-of-default-relevant signal: declining utilization, rising faults) and LGD (loss-given-default-relevant: depreciated value × condition multiplier) are kept as two separate outputs, never blended. Known gaps, stated honestly: no lifetime/CECL-horizon projection yet, no portfolio-level concentration-risk view yet, moral-hazard/gaming-risk tampering checks not built, bias/fairness testing not built, model-governance documentation not written.

THE APR BRIDGE — BOTLIEN NEVER OUTPUTS THE RATE ITSELF

Expected Loss = PD × LGD × EAD, feeding a lender's own APR = cost of funds + expected loss + capital charge + servicing cost + margin. Cost of funds, capital charge, and margin stay the lender's own proprietary economics. This scope boundary is deliberate and stays firm regardless of who the primary customer is.

4 — Operator dashboard + alerting

A simple web view per RaaS operator showing their whole fleet, current status, financial-telemetry KPIs, and a flag list. Email/SMS alerts on flag triggers.

Data access model — the moat, reframed

The moat used to rest entirely on data access being a loan condition a lender required. With the RaaS operator as the primary, direct customer, that legal lock-in isn't the Day 1 moat anymore, access is operator-authorized and voluntary, like any SaaS tool connecting to a company's own systems.

DAY 1 MOAT

Workflow embedding once an operator relies on Botlien for monthly quota and cash-flow visibility, plus modeling quality that's genuinely hard to replicate without the same accumulated telemetry history.

EXPANSION-LAYER MOAT

Once real telemetry history exists and a lender is pulled in downstream, that access becomes contractual again, a loan or lease covenant, and the original legal lock-in argument re-applies.

Flagged honestly: this is a real trade, not a strict improvement over the old lender-first moat, worth revisiting once there's real usage data to see how sticky the product proves to be on its own.

Hard scope line: Botlien only ever touches non-biometric, non-interaction telemetry, uptime, location, faults, utilization, task-completion events. Never voice, video, or anything involving human interaction data, stated explicitly in every agreement.

Regulatory grounding

Relevant primarily once the downstream lender layer activates, not a Day 1 constraint with a RaaS operator as the direct customer.

BASIS	RISK	WHY IT HOLDS
UCC Article 9	LOW	Lenders already required to periodically confirm collateral exists. Relevant once a lender is added downstream.
CECL (ASC 326)	LOW	Requires a forward-looking, lifetime loss estimate. Botlien's cash-flow-projection output feeds this directly once a lender is involved.
SR 11-7	MEDIUM	Only applies once a bank uses Botlien's output as a vendor model. Not a Day 1 concern with a RaaS operator as the direct customer.
ECOA / Regulation B	MEDIUM-HIGH	Only applies if the RaaS operator itself uses Botlien's output to make credit/leasing decisions about its own downstream customers. Secondary exposure.
FCRA	LOW	Governs consumer credit reporting. Botlien's customers are commercial entities, likely out of scope, conditional on staying commercial-only.

Not legal advice, this is reasoned analysis for planning purposes. Real counsel review is still required before any consent or covenant language goes live.

Go-to-market sequencing

RaaS operators first and foremost, lenders as a downstream expansion once there's real usage data, robot manufacturers (OEMs) after both. A RaaS operator owns its fleet and carries risk directly on every unit it leases out, one buyer in the room, no loan-closing dependency to sell into. It resolves "who pays" outright, and it's selling into the faster-growing market: RaaS at ~20% CAGR versus equipment finance's 10-13%.

OEM adverse selection is still the reason OEMs come last: participation itself becomes a signal, a manufacturer with reliable robots wants in, one with a rougher record has the opposite incentive. Cross-brand benchmarking stays an explicit opt-in product built later.

Expansion path: RaaS operators (paying, primary customer) → additional RaaS operators via case study → that operator's own upstream lender, as a verified-data upsell once real usage history exists → OEMs (as data partners, and direct customers where they self-finance) → traditional asset-backed equipment lenders, now the last stop rather than an early target.

Monetization

SEGMENT	MODEL
Small/independent RaaS operators	Per-robot, \$100/month baseline (\$1,200/yr). Tiered refinement by robot category still being finalized (\$40 delivery / \$75 warehouse AMR / \$100 hospitality-service / \$150 humanoid, pending final lock). Anchored above Samsara's fleet telematics pricing (\$27–60+/month) since Botlien includes financial-telemetry and cash-flow analytics, not just raw monitoring.
Enterprise RaaS accounts	Flat annual contract with a true-up clause at renewal, standard practice for usage-based SaaS at scale (Twilio, Datadog, Snowflake).
Lenders (downstream, secondary)	Once an operator has real telemetry history, offering verified fleet data to their upstream lender becomes a natural upsell, sold as a premium add-on, not the entry motion.

Why a RaaS operator actually pays

REASON	TYPE
Know instantly if a fleet is falling short of monthly quota or contract target, before it becomes a cash-flow surprise at month-end	Operational visibility
Cleaner cash-flow forecasting and sharper reporting to their own investors, board, or lender	Planning and reporting
Verified, real-time fleet performance history makes them a more fundable counterparty when raising their own capital	Capital access

"Who pays" is resolved by the pivot itself: the RaaS operator pays for their own tool, the same way any business buys its own analytics software.

Market sizing — TAM / SAM / SOM

Rebuilt 2026-08-05 for the RaaS-operator-primary positioning, replacing the old lender-facing \$177M figure. "RaaS operator" is a broader buyer population than "financed/leased robot" was.

INPUT	VALUE	STATUS
Global service robot installed base	475K–710K robots	Sourced
RaaS market size, 2025	\$26.7B–\$33.6B high cluster	Sourced, firms diverge 5–10x
RaaS market CAGR through 2030	~18–21%	Sourced
Seegrid: 2025 contracts that were RaaS-structured	70%	Sourced, single vendor
RaaS share of installed base	30–50%, mid 40%	Assumption
Software/analytics attach rate	~1% of RaaS revenue	Assumption
Fee per robot/yr	\$1,200	Existing anchor
TAM, bottom-up (robots × fee)	~\$283M	Derived
TAM, top-down (market \$ × attach rate)	~\$300M	Derived
TAM, central estimate	~\$290M/yr (range ~\$85M–\$500M+)	Derived
SAM, near-term reachable segment	~\$70–75M/yr	Assumption, 25% of TAM
SOM, realistic 3-yr capture	~\$0.7–2.25M ARR	Derived, 1–3% of SAM

Two independent methods, counting robots and taking a share of total RaaS dollars, converge on roughly the same ~\$280–300M range, real cross-validation, though both rest on stated assumptions rather than fully independently sourced figures, treat the convergence as reassuring, not as proof of precision.

Average RaaS company revenue, projected

Almost no pure-play RaaS companies exist yet at meaningful scale, so this is a projection built on real anchor points, not a lookup from a clean dataset.

REAL NAMED ANCHORS, TODAY

Seegrid, \$70.6M revenue. **Rapyuta Robotics**, \$38M revenue. Both mature, well-funded players, likely well above the median operator, not representative of the typical company at this stage.

THE MORE TYPICAL PROFILE

Botlien's own discovery contacts, Formic Technologies, Gatsby, and IntBot, are early-stage operators with no disclosed revenue anywhere near Seegrid's scale, the more representative sample size for now.

The honest picture is bimodal: a handful of scaling leaders in the tens of millions, and a long tail of early operators likely under \$5M, consistent with a market this early. Projected forward on the sector's ~20% CAGR: expect average revenue per operator to stay flat or even dip over the next 2–3 years as new entrants keep forming faster than the market matures, then rise later in the decade as the category consolidates and weaker operators exit or get acquired. Standard land-grab-to-consolidation shape, not a straight-line forecast.

Team

Antonio D'Angelo

CEO & CO-FOUNDER, MAJORITY STAKEHOLDER

Business Administration, Finance concentration, Cal State Fullerton. Go-to-market experience at a private-equity-owned tech company. Private equity and venture capital background. Founder of Comprply, a shipped desktop application.

Samuel Kim

CO-FOUNDER

Business Administration, Finance concentration, Cal State Fullerton. Started in computer science before transitioning to finance driven by a passion for credit and risk analysis, the exact intersection this company needs. Building the product and dashboard.

The ask

\$500K

PRE-SEED ROUND

Use of funds framed around a milestone: one or more signed pilots with real RaaS operators, generating first monitored-robot volume and a real case study, the trigger to raise a seed round.

At a lean 2-person team's typical early-stage burn, \$500K plausibly buys roughly 12–18 months of runway. Directional planning assumption, not a confirmed budget.

Exit & strategic fit

The natural acquirer is a bank or insurer, not another SaaS company. Firms with equipment-finance arms are the most logical long-term acquirer of collateral- and cash-flow-risk data infrastructure like Botlien.

PRECEDENT	RELEVANCE
Lemonade acquires Metromile (~\$500M, all-stock, July 2022)	Strongest real comparable. A financial-services acquirer buying a telematics/usage-based-risk-data company to fold proprietary sensor data directly into underwriting.
JPMorgan acquires Aumni (~\$232M, 2021)	A major bank buying a data-analytics company outright to own proprietary insight. Proves the "bank buys data company for the insight" playbook is real.

No precedent exists of a bank specifically acquiring a collateral-valuation or equipment-finance-data company, that deal hasn't happened yet, arguably the whole opportunity. What's real and citable: WTW's 2026 financial-institutions M&A outlook names "AI, analytics, and digital platforms that improve underwriting" as an explicit acquirer priority.

Discovery & traction status

TRACK	STATUS
RaaS operators (primary)	Verified contact list built. Karl Barry, Head of Robotics Monitoring and Maintenance at Formic Technologies, confirmed as the sharpest-fit contact found across any track, email drafted, not yet sent. Formic's own upstream lender, Mitsubishi HC Capital America, is a confirmed, named data point.
Manufacturers	First outreach sent to Lei Yang, CEO & Co-Founder of IntBot, met in person, via hello@botlien.com, awaiting reply.
Lenders (secondary/downstream)	Verified contact list built, held for the downstream expansion motion. Stephen Leege, SVP & Chief Credit Officer at Financial Pacific Leasing, confirmed strongest contact when this layer activates.

This is honestly where a \$500K pre-seed round should find the company. The ask is capital to convert RaaS-operator discovery into signed pilots, not to scale an already-proven model.

Honest self-audit

Scored across 7 dimensions, three checkpoints the same week so the delta from each real event is visible. Internal use only.

DIMENSION	PRE-PIVOT	POST-PIVOT	POST-RESEARCH	WHY
Market timing	8	8.5	8.5	Now targeting the specific faster-growing segment directly
Differentiation/competition	7	6.5	7	Dipped when unresearched, recovered once Findustrial/Formant were verified as real but distinct comps
Legal/regulatory grounding	6.5	7.5	7.5	SR 11-7/ECOA no longer Day-1 concerns since the direct customer isn't a bank
Product/technical validation	3	3	3	Unchanged, no new proof exists either way
Business model clarity	5.5	8	8	"Who pays" resolved outright
Team	5	5.5	5.5	Faster SMB motion fits Antonio's shipped-product background slightly better
Traction	2	2.5	2.5	Existing RaaS contacts now sit on the primary path

~5.3 → ~5.9 → ~6.0 / 10

COMPOSITE SCORE

The pivot itself did almost all of the real work (business model clarity, legal surface); the competitive research mainly restored confidence in a score that had been honestly discounted for being unverified. Product/technical validation and traction are still what caps the whole score, both only move by building the thing and landing a real pilot.

What's still missing before this is VC/PE-ready

1. No working demo yet, no investor has anything to watch work.
2. No pilot, no completed real conversation, one outreach email sent as of this update.
3. The core signal is unproven, does financial telemetry actually predict quota shortfall or default.
4. No real financial projections beyond the illustrative TAM/SAM/SOM math, needs a quarter-by-quarter revenue ramp.
5. Samuel's technical scope for this specific build is a real, acknowledged open question.
6. No actual legal counsel review yet, everything regulatory above is reasoned analysis.
7. Whether operators' own contract/billing data is standardized enough to integrate quickly is unvalidated.

Open questions to resolve early

- Samuel's exact technical scope and robotics-specific experience, the single largest risk to execution feasibility.
- Whether Bear Robotics or Pudu will grant production commercial access to a startup for this specific use case.
- Does asset/telemetry decline actually predict default or quota shortfall, still completely unproven, the real test, not the API access question.
- Billing/contract-data integration feasibility, unknown how standardized RaaS operators' own systems actually are.
- IntBot's one determining fact, still unknown: does IntBot own and lease out its deployed robots (self-financing RaaS, natural first customer), or is hardware sold outright to the operator?

Phased build plan

PHASE	WEEKS	GOAL
0 — Foundation	1–3	Confirm a real robot brand's API actually exposes usable telemetry.
1 — Single-Brand Prototype	4–9	One robot, one brand, one operator-facing dashboard screen.
2 — Design Partner	10–18	Get one real RaaS operator using this on at least one real deployed robot.
3 — Expand + Raise	19–26	Add a second robot brand, tighten the pitch with real pilot data, start outreach.

Two-person team, part-time around school/existing jobs. ~6 months to a real design-partner pilot. Riskiest part is Phase 0, everything downstream depends on real telemetry access.

Implementation log

- ✓ botlien.com domain live, DNS on Cloudflare, info@ and hello@ both routing correctly.
- ✓ Landing page built, deployed, and updated for the RaaS-primary positioning (copy only, photos untouched).
- ✓ Bear Robotics validated as the Phase 0 technical bet.
- ✓ Risk-scoring methodology hardened across two adversarial audit passes.
- ✓ Three-track discovery tracker built and reordered, RaaS operators now Track 1/primary.
- ✓ First real outreach sent, Lei Yang (IntBot).
- ✓ Pitch deck built as a live HTML artifact, fully rewritten for RaaS-primary positioning.
- ✓ Positioning pivot: RaaS operator is now the primary customer, not the lender, all source files updated in one pass.
- ✓ Competitive research on RaaS fleet-financial-analytics, verified twice, Findustrial and Formant named directly.
- ✓ RaaS-specific TAM/SAM/SOM rebuilt (~\$290M central, cross-validated two ways), average-RaaS-company-revenue projection built from real named anchors.
- Send drafted outreach to Karl Barry (Formic Technologies) and Stephen Leege (Financial Pacific Leasing).
- Lock final tiered per-robot pricing by robot category.
- Land one real pilot conversation, the single highest-leverage open item across the entire business.
- Validate billing/contract-data integration feasibility with a real RaaS operator.
- Real legal counsel review of all regulatory reasoning in this document.